

Shehab Ahmed Taia

Mechatronics Engineering MEng Student

University of Birmingham

shehab3092006@gmail.com — +44 7918243781 — [linkedin.com/in/shehab-taia](https://www.linkedin.com/in/shehab-taia)

Education

Year 2 – MEng Mechatronics Engineering

University of Birmingham

Expected Graduation: 2028

Relevant Modules: Digital Electronics, Engineering Mathematics, Mechanics, Electrical Machines

Year 1 (Pathway) – Mechatronics Engineering

Kaplan International College London

Academic Year: 2024–2025

Relevant Modules: Engineering Mathematics, Mechanics, Thermodynamics, Thermofluids, Research Design Development, Programming

Skills

Software & Tools: MATLAB, AutoCAD, TinkerCAD, Multisim Live, Microsoft Excel, LaTeX

Embedded & Programming: C/C++(Arduino), Python

Technical: Circuit prototyping, embedded systems, debugging, technical documentation

Projects

Autonomous Fire-Fighting Vehicle

January – June 2025

Six-member team project

- Played a leading role in the technical development of an autonomous fire-fighting vehicle using an **Arduino Uno R3**
- The system was built and tested in **TinkerCAD**, simulating 13+ components
- Wrote 50+ Arduino code from scratch to control independent wheel motors, ultrasonic distance sensing, infrared fire detection, and a water pump extinguishing system
- Debugged power supply and motor control issues to ensure reliable autonomous operation
- Delivered an operating system capable of avoiding obstacles, locating a fire source, and extinguishing fires. Validated through repeated test scenarios

Digital Electronics – Hardware Finite State Machine

October – December 2025

Two-member team project

- Designed a synchronous finite state machine with 13 specified states, including handling unallocated states
- Collaboratively developed state diagrams, transition tables, and Karnaugh maps to ensure logical correctness
- Implemented and validated the logic in **Multisim Live**, followed by breadboard layout and wiring planning in **TinkerCAD**
- Built and tested the physical circuit using **7474 D flip-flops**, clock logic, and **7400/7410/7420 NAND gates**
- Documented the full design and testing process in a structured engineering logbook with 30+ pages of structured documents

Work Experience

IT Department Internship: Talaat Mostafa Group

December 2023

- Completed a short introductory placement within the IT department, gaining exposure to enterprise IT environments and day-to-day professional IT workflows
- Gained exposure to basic networking concepts, Microsoft server environments, disk management, and server room operations